Scalar-level initial geometry and robust shell-volume perturbations

Ricardo Maldonado
Independent Researcher

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Abstract

We describe a geometric, background-calibrated class of homogeneous second preparations in a conditional five-dimensional Einstein–scalar shell model. On a physically supplied initial Cauchy section, we compare the three-area of the continued spherical component of each scalar level with that of a fixed base preparation. For families with the same first tangent, the relative difference of second area derivatives is $D = 3J/2 - 3aw/p$, including the scalar displacement of the level. In the retained compact metric-mean freedom, with the other canonical sectors fixed, the full profile determines the preparation by a two-component linear initial-value problem. On the target's causal initial segment, exact rational bounds give $N \le R_A/2000$, where R_A bounds D and its fixed background scalar-label derivative. The specified class $R_A \le 2 \times 10^{33}$ therefore retains strictly negative proper-volume and expansion second coefficients at one fixed proper time. Each admitted exact family consequently has smaller positive volume and slower positive expansion for sufficiently small nonzero parameter of either sign. The result retains explicit background, constraint, development, and uniqueness premises. It supplies neither a physical rule choosing this class nor a parameter interval uniform over it. The contribution is the restricted geometric reconstruction and its quantitative transport to these two model observables, rather than a general gauge-invariance principle.

1 Question, model, and retained premises

A fixed first perturbation does not fix homogeneous second initial data. In the present shell model, an admissible compact second preparation can reverse the signs of both proper-volume and expansion coefficients while preserving the first packet and the initial shell collar [4, 5]. This raises a narrower question: can a specified class of second preparations be expressed using initial geometry, and can one certify both signs throughout that class? We answer this conditionally for the retained metric-mean freedom. The initial physical section and a background scalar-label calibration remain part of the specification.

The bulk action has normalization

$$S_{\mathrm{bulk}} = \int \left(\tfrac{1}{2}R - \tfrac{1}{2}|\mathrm{d}\phi|_g^2 - \mathcal{V}(\phi)\right) \mathrm{d}\mu_g. \tag{1}$$

The bulk and one-copy outward shell equations are

$$\mathrm{Ric}_{AB} - \phi_A\phi_B - \tfrac{2}{3}\mathcal{V}(\phi)g_{AB} = 0, \qquad \Box_g\phi - \mathcal{V}'(\phi) = 0, \tag{2}$$

$$K^{\mathrm{out}}_{ab} = \tfrac{1}{6}\sigma(\phi)\gamma_{ab}, \qquad n_{\mathrm{out}}\phi = -\tfrac{1}{2}\sigma'(\phi). \tag{3}$$

There are two identical interior copies with their respective outward normals. Fix

$$W(\phi) = 1 - \phi + \phi^3/3, \qquad \mathcal{V} = \tfrac{1}{2}(W')^2 - \tfrac{2}{3}W^2, \qquad \sigma = 2W + \delta_\sigma(1 + c_\sigma\phi),$$
$$\delta_\sigma = 10^{-3}, \qquad c_\sigma = 96714321425692/161839258930731. \tag{4}$$

The selected regular near-$\phi = 1$ background is

$$g_0 = \mathrm{d}y^2 + \rho(y)^2[-\mathrm{d}t^2 + \cosh^2 t\,\kappa], \qquad a = \rho_y/\rho > 0, \quad p = \phi_{0,y} \neq 0, \tag{5}$$

where κ is the unit round metric on S^3. Its exact selected shell radius is ρ_b. Put $\ell = 10^{-6}$, $x = t/\ell$, $r = z/\ell$, and $\mathrm{d}y = -\rho\,\mathrm{d}z$. The packet is $1 \le r \le 2$; the target is

$$T = 5/4, \qquad \tau_* = \rho_b \ell T. \tag{6}$$

The exact ρ_b, rather than a rounded value from an enclosure, specifies this clock. All quantities use the inherited model and coefficient units.

Every comparison keeps the selected background, first seed and parameter normalization, full quadratic constraint and evolution sources, canonical momentum and scalar canonical second sectors, second transverse-traceless initial pair, and initial shell collar fixed. In particular,

$$x_\varepsilon = x_0 + \varepsilon v_{\mathrm{seed}} + \tfrac{1}{2}\varepsilon^2 x_2 + o(\varepsilon^2), \qquad g_\varepsilon = g_0 + \varepsilon q + \tfrac{1}{2}\varepsilon^2 k + o(\varepsilon^2). \tag{7}$$

The fixed first seed is a smooth flat quadrupole packet with $Y = n_1^2 - n_2^2$, $\langle Y\rangle = 0$, $\langle Y^2\rangle = 1/6$, and $\Delta_\kappa Y = -8Y$. Its normalization and the base preparation are identified in Appendix A. Its full first-field solution is common to all families; no quadratic first-field term is discarded in an absolute observable.

We retain the selected background and its continuous authenticated annulus bounds; the equivariant weighted constraint isomorphism and nonlinear implicit-function hypotheses; the conditional fixed-shell development, subsidiary and boundary uniqueness, and identification of the first derivative with that first-field solution. For each fixed preparation we assume C^2 dependence on ε in the retained lower C^4 spacetime topology on a common compact neighborhood, including the required traces. After a family-dependent shrinkage in ε, an intrinsic normal tube reaches (6). The complete five-mean linear mixed problem retains its physical Robin and normal-gauge rows, uniqueness, and the coefficient estimate stated in Section 6. These are shared analytic premises of the retained construction [5]; the finite rational checks below do not establish them. No effective or uniform nonlinear parameter radius is assumed.

2 Restricted homogeneous preparation freedom

On the physical initial section Σ, the background spatial data are $\gamma_0 = \mathrm{d}y^2 + \rho^2\kappa$ and $\phi_0 = \phi_0(y)$. Let h be smooth, supported in the packet and flat at its endpoints, with

$$\int_{y_L}^{y_U} \rho^2 h\,\mathrm{d}y = 0, \quad \chi(y) = \int_{y_L}^{y} \rho^2 h\,\mathrm{d}s, \quad h = \rho^{-2}\chi_y, \qquad y_L = y(2), \quad y_U = y(1). \tag{8}$$

Thus χ is smooth, compact, and flat. The complete added second preparation is the metric mean

$$\Delta\gamma_2 = h\gamma_0 + \mathcal{L}_{\zeta\partial_y}\gamma_0, \quad \Delta\phi_2 = 0, \qquad \zeta = \frac{3(h_y - ah)}{2p^2} + \frac{9\chi}{2\rho^4 p^2}; \tag{9}$$

the canonical momentum and both scalar canonical second sectors are fixed. The background identity $a_y + 4a^2 = 3/\rho^2 - 2\mathcal{V}/3$ gives

$$\begin{aligned} C[h] &= -\tfrac{3}{2}(h_{yy} + 3ah_y) - \mathcal{V}h, \\ \rho^4 C[h] &= -\tfrac{3}{2}[\rho^4(h_y - ah)]_y - \tfrac{9}{2}\rho^2 h, \qquad\qquad P_g\Delta\gamma_2 = C[h] + \rho^{-4}(\rho^4 p^2\zeta)_y = 0. \end{aligned} \tag{10}$$

The moment and flatness conditions make this a compact homogeneous constraint datum. Adding it to the base preparation leaves the complete quadratic constraint source unchanged. The retained weighted implicit-function construction and equivariance give an exact completed family with the stated second mean for each fixed admissible h [4]. This is an application of the premises, not a new existence theorem for unrestricted data.

The freedom is not the pure second conditioning diffeomorphism. At the time-symmetric background, a normal generator changes the initial spatial metric and scalar by multiples of $K_0 = 0$ and $\Pi_0 = 0$. Averaging a proposed tangential generator leaves a radial one; its scalar variation is pX^y. Since the raw scalar difference in (9) vanishes and $p \neq 0$, $X^y = 0$. Also $\Delta\gamma_2 = 0$ would require $h + 2\zeta_y = h + 2a\zeta = 0$, hence $\zeta_y = a\zeta$; compact support then gives $\zeta = h = 0$. Nonzero h is therefore physical in the retained anchored pure second gauge quotient, with the first representative fixed.

3 Scalar-level area on the supplied section

For a fixed background scalar value $\varphi = \phi_0(y)$, let $S_\varepsilon(\varphi)$ be the component of $\{\phi_\varepsilon = \varphi\}$ in the selected annular collar that continues the background three-sphere. Define

$$A_\varepsilon(\varphi) = \mathrm{Vol}_3(S_\varepsilon(\varphi), \gamma_\varepsilon), \qquad A_0(\phi_0(y)) = 2\pi^2\rho(y)^3, \qquad D_h(y) = \frac{\partial_\varepsilon^2 A_\varepsilon^h(\phi_0(y))|_0 - \partial_\varepsilon^2 A_\varepsilon^{\mathrm{base}}(\phi_0(y))|_0}{A_0(\phi_0(y))}. \tag{11}$$

The scalar label is held fixed in the parameter derivatives. They are actual second derivatives, with no extra factor 1/2. The component qualification excludes unrelated components elsewhere on Σ.

Lemma 1 (Regularity and shared-first cancellation)**.** *Under the retained C^2-into-C^4 hypothesis, (11) exists and is C^1 in the background label. If two preparations have identical first metric and scalar fields, then*

$$D_h = \langle \tfrac{1}{2}\,\mathrm{tr}_{\rho^2\kappa}(\Delta\gamma_{2,\tan}) - 3a\Delta\phi_2/p \rangle. \tag{12}$$

Brackets denote the normalized κ integral.

Proof. Work in a slightly enlarged compact collar, so endpoint labels are included. The uniform nonzero bound on p and C^4 closeness give unique continued level graphs $F_\varepsilon(y,\omega) = (u_\varepsilon(y,\omega),\omega)$ satisfying

$$\phi_\varepsilon(u_\varepsilon(y,\omega),\omega) = \phi_0(y), \qquad u_0 = y. \tag{13}$$

The spatial implicit equations give continuity of u_ε in C^4. At the graph, its parameter derivatives satisfy

$$\dot{u} = -\dot{\phi}/\phi_y, \qquad \ddot{u} = -\frac{\ddot{\phi} + 2\dot{\phi}_y\dot{u} + \phi_{yy}\dot{u}^2}{\phi_y}. \tag{14}$$

Their spatial derivatives through order two use at most four spatial derivatives of the input and uniformly nonzero denominators. Difference quotients and the mean value formula give u_ε C^2 into C^2. This allows two spatial orders of loss; differentiability of composition at unchanged C^4 regularity is not needed.

The induced metric, with all fields evaluated at F_ε, is

$$\sigma_{\varepsilon,AB} = \gamma_{\varepsilon,AB} + \gamma_{\varepsilon,yA}\partial_B u_\varepsilon + \gamma_{\varepsilon,yB}\partial_A u_\varepsilon + \gamma_{\varepsilon,yy}\partial_A u_\varepsilon \partial_B u_\varepsilon. \tag{15}$$

It is C^2 into C^1 and remains positive definite. Integrating $\sqrt{\det \sigma_\varepsilon}$ over the compact sphere gives A_ε C^2 into C^1. No rotational symmetry of the perturbed data is assumed.

The first level displacements coincide. Subtracting (14) at $\varepsilon = 0$ gives $\Delta\ddot{u} = -\Delta\phi_2/p$. The background angular graph slope and $\gamma_{0,yA}$ vanish, so

$$\Delta\sigma_{2,AB} = \Delta\gamma_{2,AB} - (\Delta\phi_2/p)\partial_y\gamma_{0,AB}. \tag{16}$$

The determinant terms quadratic in the common first variation cancel after subtraction. Its remaining linear trace gives (12). The actual nonspherical first-field quadratics remain in each absolute area variation. □

For an angular metric mean J and scalar mean w, the formula becomes

$$\boxed{D_h = \tfrac{3}{2}J - 3aw/p.} \tag{17}$$

In (9), $J = h + 2a\zeta$ and $w = 0$. After the retained pure second conditioning gauge, $J = h$ and $w = -p\zeta$. Both give

$$D_h = \tfrac{3}{2}(h + 2a\zeta), \qquad w = \frac{p}{2a}(h - 2D_h/3). \tag{18}$$

The scalar pullback in the conditioned representative is essential. For a general pure second tangential generator X, the metric trace in (12) contributes $3aX^y + \mathrm{div}_\kappa X^{\tan}$ and the scalar term contributes $-3aX^y$. The sphere integral of the divergence vanishes. Thus the area is invariant under section diffeomorphisms transporting both fields, including nonspherical ones. A spacetime diffeomorphism must also transport the supplied physical section. Replacing it by a different physical slice changes the observable; no invariance under that replacement is asserted.

4 Reconstruction and the geometric class

Set $Z = \chi/(\ell\rho^3)$, $b = \ell\rho a$, and $g = \ell\rho p^2/(3a)$. Substitution of (9) into (18) gives

$$h_y = (a - p^2/(3a))h - 3\chi/\rho^4 + (p^2/(3a))(2D_h/3). \tag{19}$$

Using $\rho_r = -\ell a\rho^2$ and $\chi_r = -\ell\rho^3 h$, we obtain

$$\boxed{Z_r = 3bZ - h, \qquad h_r = 3\ell^2 Z + (-b+g)h - \tfrac{2}{3}gD_h, \qquad (Z,h)(1) = (0,0).} \tag{20}$$

Flatness gives $h(1) = 0$ and the exact zero moment gives $\chi(1) = 0$.

Proposition 1 (Restricted injectivity)**.** *Within* (8)–(9)*, with the other canonical second sectors fixed, equal profiles on* $[1,T]$ *give equal* h*,* χ*, and* ζ *on that segment. Equal full profiles on* $[1,2]$ *give equal full preparations. In particular,* $D_h = 0$ *on the whole packet implies* $h = 0$*.*

Proof. The continuous-coefficient linear initial-value problem (20) has a unique solution by its fundamental matrix or the integral Gronwall inequality. Its difference for equal profiles has zero forcing and initial value. The formula for ζ then gives the claim. □

A profile zero only on $[1,T]$ need not restrict a preparation outside that causal segment. This is injectivity for the specified mean freedom, not for arbitrary Einstein–scalar Cauchy data. It is not surjectivity onto arbitrary C^1 profiles: compactness, endpoint flatness, and the moment condition constrain the image.

The derivative used to calibrate the class is fixed by the background scalar label:

$$D_{h,r} := \phi_{0,r}\partial_\varphi D_h = (-\ell\rho p)\partial_\varphi D_h, \qquad \varphi = \phi_0(y(r)). \tag{21}$$

It is not a derivative in an arbitrary perturbed radial coordinate. The background calibration can itself be read from $\rho(\varphi) = (A_0(\varphi)/(2\pi^2))^{1/3}$ and the retained map $r \mapsto \varphi$. Define

$$R_A(h) = \max\left\{\sup_{1\le r\le T}|D_h(r)|, \sup_{1\le r\le T}|D_{h,r}(r)|\right\}, \qquad \mathcal{C}_A = \{h \text{ satisfying (8)} : R_A(h) \le 2\times 10^{33}\}. \tag{22}$$

This is a geometric, background-calibrated specification on the given section. Neither the restriction to these canonical sectors nor the chosen radius is derived from a physical preparation process. If an increment is written λh, its class norm is $R_A(\lambda h) = |\lambda| R_A(h)$; the unscaled direction does not control arbitrary λ.

5 An exact rational initial-data bound

Write $R = R_A(h)$. The retained continuous background boxes imply, everywhere on the causal initial segment,

$$\begin{aligned} &129 \le \rho, \rho_b \le 131, \qquad 3/25000 \le -p \le 1/7000, \qquad 111/1000 \le a \le 14/125, \\ &|p_y| \le 1/4000, \qquad |a_y| \le 1/10000, \qquad 1/1200 \le -F_b' \le 1/1000. \end{aligned} \tag{23}$$

Here F_b' is the scalar-field derivative at the background shell of $F(\phi) = 2\mathcal{V}(\phi) - \sigma'(\phi)^2/4 + \sigma(\phi)^2/3$, the coefficient in the retained proper-volume difference equation below. These are bounds on continuous domains, rather than point samples. Their source bytes, the class radius, and the numerical gates were fixed before the new rational evaluation.

Let $b_+ = 917/62500000$ and $g_+ = 131/16317000000000$, obtained directly from (23). The matrix of (20) has infinity norm at most

$$\max\{1 + 3b_+, 3\ell^2 + b_+ + g_+\} \le \mu := 20001/20000. \tag{24}$$

Put $\Delta = T - 1 = 1/4$. A positive Taylor sum through degree six with a geometric tail gives the exact rational certificate

$$e^{\mu\Delta} \le \sum_{j=0}^{6} \frac{(\mu\Delta)^j}{j!} + \frac{(\mu\Delta)^7/7!}{1-\mu\Delta/8} < \frac{4}{3}. \tag{25}$$

Variation of constants therefore bounds both $|Z|$ and $|h|$ by HR, where

$$H = \Delta\tfrac{4}{3}\tfrac{2}{3}g_+ = \frac{131}{73426500000000}, \qquad H_r = 3\ell^2 H + (b_+ + g_+)H + \tfrac{2}{3}g_+, \quad |h_r| \le H_r R. \tag{26}$$

No derivative bound on the ODE matrix is required.

In (K, L, N, J, w) order the conditioned difference data satisfy

$$W_0 = (h, 0, 0, h, w), \qquad W_{0,x} = (0, V_L, 0, 0, 0), \qquad W_{0,r} = (h_r, 0, 0, h_r, w_r), \tag{27}$$

where, with $k = p/(2a)$,

$$w = k(h - 2D_h/3), \qquad w_r = k_r(h - 2D_h/3) + k(h_r - 2D_{h,r}/3), \qquad V_L = h_r + bh. \tag{28}$$

The mixed metric velocity and the derivative of k are both retained. Bounds following from (23) are

$$\begin{aligned} k_+ &= \frac{1/7000}{2(111/1000)} = \frac{1}{1554}, \\ k_{r,+} &= \ell\, 131 \left\{ \frac{1/4000}{2(111/1000)} + \frac{(1/7000)(1/10000)}{2(111/1000)^2} \right\}, \\ S &= k_+(H + 2/3), \qquad S_r = k_{r,+}(H + 2/3) + k_+(H_r + 2/3), \qquad V = H_r + b_+ H. \end{aligned} \tag{29}$$

Thus $|w| \le SR$, $|w_r| \le S_r R$, and $|V_L| \le VR$. For $\mathsf{H} = \mathrm{diag}(1, 2, 1, 3, 1)$, let $I_0 = \sup |W_0|_{\mathsf{H}}$, $I_r = \sup |W_{0,r}|_{\mathsf{H}}$, $I_x = \sup |W_{0,x}|_{\mathsf{H}}$, with the data zero for $r \le 1$. The triangle inequalities give

$$I_0 \le (2H + S)R, \qquad I_r \le (2H_r + S_r)R, \qquad I_x \le 2VR, \qquad N := \max\{(I_0 + TI_x)/3, I_r + I_x\}. \tag{30}$$

All operations are rational. The resulting exact upper bound is

$$\begin{gathered} N \le c_N R \le R/2000, \\ c_N = \frac{7989167107795594661759312030 77}{18618477115770000000000000000000000} < 0.000429098849392 < \frac{1}{2000}. \end{gathered} \tag{31}$$

In particular every member of (22) satisfies $N \le 10^{30}$. The certificate verifies 38 finite checks using standard-library rational arithmetic, with zero new background or evolved-field oracle calls, PDE solves, or bump queries. A separate rational implementation reconstructs the bound from shared retained inputs without importing the producer [5]. These checks share the analytic and background premises.

6 Transport to proper volume and expansion

The shell observables use the intrinsic future normal-geodesic foliation anchored at the common initial section:

$$V_\varepsilon(\tau) = \int_{S^3} \mathrm{d}\mu_{h_\varepsilon(\tau)}, \qquad \Theta_\varepsilon = \partial_\tau \log V_\varepsilon, \qquad v_2 = \frac{\partial_\varepsilon^2 V_\varepsilon|_0}{V_0}, \quad \Theta_2 = \partial_\varepsilon^2 \Theta_\varepsilon|_0. \tag{32}$$

The clock and anchor are transported under gauge changes. The background has $A(\tau) = \rho_b \cosh(\tau/\rho_b)$, $V_0 = 2\pi^2 A^3$, and $\Theta_0 = 3\rho_b^{-1} \tanh(\tau/\rho_b)$.

The same first field means the same quadratic bulk, shell, clock, and measure sources. They cancel in the difference of the second responses. Retained linear uniqueness gives the complete homogeneous five-mean difference W, including metric–scalar coupling and Robin feedback. On the backward-closed cone $0 \le x \le T$, $0 \le r \le T - x$, the retained coefficient estimate has

$$q = \frac{1201}{3200000} < 1, \qquad X(U) = \max\{\sup |U|_{\mathsf{H}}/3, \sup |U_x|_{\mathsf{H}}, \sup |U_r|_{\mathsf{H}}\}. \tag{33}$$

For the free reflected wave B of (27), $X(B) \le N$ and $X(W - B) \le qN/(1-q)$, hence $X(W) \le N/(1-q)$. The exact zero wedge $x + r \le 1$ then implies, for $m(x) = W_w(x, 0)$,

$$|m(x)| \le (x - 1)_+ \frac{N}{1 - q}. \tag{34}$$

Only the initial segment up to T enters this target estimate.

Let $\nu = v_2^h - v_2^{\mathrm{base}}$ and $\theta = \Theta_2^h - \Theta_2^{\mathrm{base}}$. The retained proper-time difference equation and zero initial conditions are

$$\nu_{\tau\tau} + 4(A_\tau/A)\nu_\tau - 2\nu/A^2 = (F_b'/2)m, \qquad \nu(0) = \nu_\tau(0) = 0, \qquad \theta = \nu_\tau. \tag{35}$$

There is no additional $1/2$ multiplying ν or θ: the observable coefficients are second parameter derivatives. At the target,

$$\begin{aligned} \nu(T) &= (\rho_b\ell)^2(F_b'/2)\int_1^T k(T,x)m(x)\,\mathrm{d}x, \\ \theta(T) &= \rho_b\ell(F_b'/2)\int_1^T k_T(T,x)m(x)\,\mathrm{d}x, \\ k(T,x) &= \frac{\sinh(2\ell(T-x))}{2\ell}\frac{\cosh^2(\ell x)}{\cosh^2(\ell T)}, \qquad k_T = \frac{\cosh^2(\ell x)\cosh(\ell T - 2\ell x)}{\cosh^3(\ell T)}. \end{aligned} \tag{36}$$

The exact inequalities $0 \le k \le T - x$ and $0 < k_T \le 1$, together with (34), give

$$\begin{gathered} |\nu(T)| \le C_v N, \qquad |\theta(T)| \le C_\Theta N, \\ C_v = \frac{(131\ell)^2(1/1000)\Delta^3}{12(1-q)} = \frac{17161}{767711760000000000}, \qquad C_\Theta = \frac{131\ell(1/1000)\Delta^2}{4(1-q)} = \frac{131}{63975980000}. \end{gathered} \tag{37}$$

This bounds the full coupled response, rather than a free-history sign.

Theorem 1 (Conditional robustness of the specified class)**.** *Under the retained premises, every admitted exact family with complete increment $h \in \mathcal{C}_A$ has, at (6),*

$$v_2 \in [-7.6900 \times 10^{16}, -4.1600 \times 10^{15}], \qquad \Theta_2 \in [-1.1070 \times 10^{22}, -2.3980 \times 10^{21}]. \tag{38}$$

For each such family there exists a possibly family-dependent $\varepsilon_h > 0$ such that, for $0 < |\varepsilon| < \varepsilon_h$ of either sign,

$$0 < V_\varepsilon(\tau_*) < V_0(\tau_*), \qquad 0 < \Theta_\varepsilon(\tau_*) < \Theta_0(\tau_*). \tag{39}$$

No positive parameter radius common to the whole class is asserted.

Proof. The retained base coefficient enclosures and the absolute difference bounds at $N = 10^{30}$ are

Quantity	Base outward enclosure	Absolute increment cap
v_2	$[-5.4520, -2.6521]\,10^{16}$	$C_v 10^{30} < 2.23535\,10^{16}$
Θ_2	$[-9.0208, -4.4458]\,10^{21}$	$C_\Theta 10^{30} < 2.04765\,10^{21}$

The exact base rational intervals plus and minus the exact caps give (38) with outward rounding. Equivalently, 10^{30} is strictly below the retained sufficient threshold $N_{\mathrm{crit}} > 1.1864 \times 10^{30}$ for both coefficients to stay negative. The first integrated variations vanish by the common quadrupole mean. For each fixed admitted family, C^2 dependence gives

$$V_\varepsilon/V_0 = 1 + \tfrac{1}{2}\varepsilon^2 v_2 + o(\varepsilon^2), \qquad \Theta_\varepsilon = \Theta_0 + \tfrac{1}{2}\varepsilon^2\Theta_2 + o(\varepsilon^2). \tag{40}$$

Since $V_0 > 0$ and $\Theta_0(\tau_*) > 0$, strict negativity gives the upper inequalities and continuity gives positivity after a further family-dependent shrinkage. This proves (39). □

The small-parameter conclusion is qualitative. The class bounds no higher norms or nonlinear remainders uniformly, so the common coefficient enclosures do not provide a uniform finite-ε regime. They concern one registered proper time, not a signed time interval, contraction, or the sign of volume acceleration.

7 The earlier reversal and the scope of the result

The earlier sign reversal uses the fixed admissible direction

$$Q(r) = \exp\!\left(4 - \frac{1}{(r-1)(2-r)}\right) \ (1 < r < 2), \qquad Q = 0 \text{ otherwise},$$
$$\chi = \ell \rho_b^3 Q, \qquad h_* = -\left(\rho_b/\rho\right)^3 Q_r, \tag{41}$$

scaled by the fixed finite $\lambda = -10^{21}$. Its two proper-response coefficients are positive in that retained construction [4]. This does not contradict Theorem 1. Independent exact rational target bounds give

$$D_{h_*}(T) > 3.7621 \times 10^{12}, \qquad D_{h_*,r}(T) < -6.870 \times 10^{13}. \tag{42}$$

Consequently

$$R_A(-10^{21} h_*) > 3.7621 \times 10^{33} > 2 \times 10^{33}, \qquad |D_{-10^{21} h_*,r}(T)| > 6.870 \times 10^{34}. \tag{43}$$

It is outside the specified class already at one endpoint. This exclusion uses retained target background boxes and an independent analytic Taylor enclosure for the old $Q(T)$ and its derivatives; it is a check of an existing witness, not a new direction search or oracle call.

The geometric ingredients have established antecedents. General first and second variations of induced hypersurface geometry are treated by Mars [1, Secs. 3, 6–7]; the cancellation in Lemma 1 is an application with a common first tangent. Scalar–metric invariant combinations in warped scalar backgrounds are standard [2, Secs. III–IV]: with transverse conformal coefficient $J = 2\psi$, our $D = 3(\psi - aw/p)$ has the same kinematic cancellation. This does not identify the two dynamical problems. The fact that a fixed scalar-defined window can give a gauge-invariant averaging prescription does not select that window or a physical hypersurface [3, Sec. 3, Eq. (3.23), and Sec. 4]. That work uses a timelike spacetime scalar gradient; here spatial scalar levels lie inside an already supplied Cauchy section. Accordingly, the result here is not a new gauge principle. It identifies the retained mean freedom through a particular level-area profile, proves its restricted reconstruction, and transports a stated profile radius to simultaneous coefficient bounds in this model.

The supplied Cauchy section, continued level component, background label calibration, canonical-sector restriction, and class radius are explicit inputs. No physical process selecting them is given. The larger freedom still admits the earlier opposite-sign preparation; the class theorem states what follows after a further specification, not sign universality from the first packet alone. No SI calibration, measured amplitude, radiation epoch, or cosmological origin follows from these model-unit coefficients. This focused comparison does not establish priority or exhaustive literature novelty. The conditional mathematical result is intended for author and specialist review.

A Identification of the fixed seed and base

For clarity, the retained seed can be specified without choosing a perturbed foliation. Let Q be (41), $B(z) = Q(z/\ell)$, p_b the background shell scalar derivative, and

$$\alpha = (\rho/\rho_b)^{3/2} p/p_b, \qquad \varkappa = \alpha_z/\alpha = \rho(5a/2 - \mathcal{V}'/p),$$
$$u = \alpha B/\rho^2, \quad u_t = e^{-z} u, \quad s_1 = u_t/\rho, \qquad v = 3\alpha(B_z + \varkappa B)/(\rho^3 p),$$
$$v_t = 3\alpha e^{-z}[B_z + (\varkappa - 1)B]/(\rho^3 p), \qquad j = v_t/\rho. \tag{44}$$

In a background orthonormal spatial frame, with spatial density ρ_0, the canonical first datum is

$$\gamma_1 = \mathrm{diag}(-4u, 2u, 2u, 2u)Y, \quad \pi_1/\rho_0 = \mathrm{diag}(-3s_1/2, 0, 0, 0)Y, \quad \phi_1 = vY, \quad \varpi_1/\rho_0 = jY. \tag{45}$$

The mean Hamiltonian and radial momentum Hessians in convention (7) are

$$\begin{aligned} Q_H = & -2u_y^2 - 24auu_y - 2uu_{yy} - (16a_y + 32a^2 + 8p^2/3)u^2 - 4u^2/\rho^2 - s_1^2 \\ & - (j^2 + v_y^2 + \mathcal{V}''v^2)/6 - (4/3)puv_y - (4/3)v^2/\rho^2, \\ Q_M = & -2s_1u_y + j(v_y + 4pu)/3. \end{aligned} \tag{46}$$

Define $I = \int_{y_L}^{y_U} \rho^4 Q_H \,\mathrm{d}y$, $J_B = \int_{y_L}^{y_U} \rho^2 B \,\mathrm{d}y > 0$, $f = 2IB/(9J_B)$, and $\xi = -(\rho^4 p^2)^{-1} \int_{y_L}^{y} \rho^4 (Q_H + C[f]) \,\mathrm{d}s$, with C as in (10). The base second mean, in $(\gamma, \pi, \phi, \varpi)$ order, is

$$B_{\mathrm{base}} = \left(f\gamma_0 + \mathcal{L}_{\xi\partial_y}\gamma_0,\ \rho_0 \,\mathrm{diag}(0, -Q_M/(6a), -Q_M/(6a), -Q_M/(6a)),\ 0,\ 0 \right). \tag{47}$$

It satisfies the full mean second constraint with these Hessians. The retained exact completion and evolution, rather than just the initial Hessians, supply the base response intervals used in the theorem. The added datum (9) leaves all canonical entries except the metric mean unchanged.

B Reproducibility and shared inputs

The accompanying capsule [5] contains the geometric proof, frozen design, rational producer, normal and optimized reports, independent reconstruction, and the previous-witness exclusion check. The rational producer authenticates the retained static annulus jets, base proper response, and preparation-difference records by SHA-256 before evaluating the fixed gates. Normal and optimized runs agree in their report bytes. The independent checker imports no producer. Its agreement checks finite arithmetic under shared background and analytic assumptions, not an independent observation or proof of the Einstein–scalar PDE.

The scalar-level geometry and reconstruction require no new background solution, evolved-field evaluation, numerical quadrature, or bump evaluation. All new class bounds are rational consequences of (23), (20), and the retained coupled estimate. The supplementary old-witness check constructs analytic target exponential bounds and does not query a numerical bump oracle. Source, displayed-bound audit, deterministic compilation record, and page-render review accompany this manuscript.

References

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